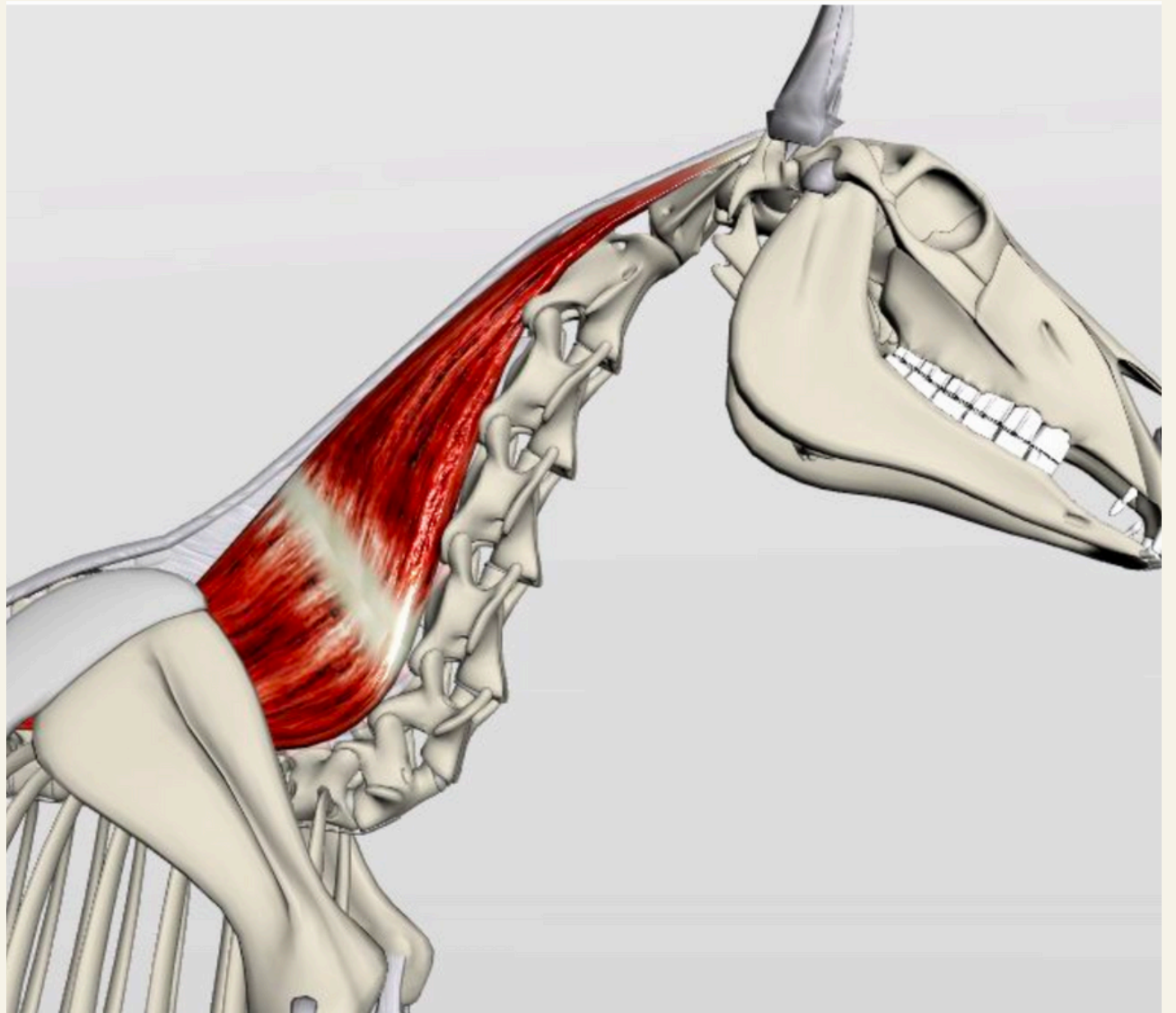


APPLIED EQUINE ANATOMY

Thoracic Sling

Thoracic Sling

The thoracic sling encompasses the shoulder joint, lower half of the neck, thoracic spine, ribcage, and sternum. Since horses do not have a clavicle (collarbone), the stability of this region during motion relies heavily on interconnected slings of soft tissues.



In general, the soft tissue chains form two “V” shapes:

- One upright V has its origin medially and distally, with the apex located between the chest, lower ribcage, and shoulder/elbow joints. The arms of this V reach proximally, anchoring onto the horse’s body in the back, barrel, and neck.
- The other is an upside-down V, with its origin on the dorsal and lateral surfaces of the withers, shoulders, and ribcage. Its arms extend downward toward the scapula, neck, and mid ribs.

The primary motion of this region involves lifting the ribcage and sternum up between the scapulae, along with limb movement and flexion/extension of the neck.

While this is not a perfect anatomical description, it effectively conveys the concept of two interconnected soft tissue slings that reciprocally lift and support the enormous weight of the forehand during motion. **This concept is explored in much greater depth in Equine Institute’s Biomechanics course.**



APPLIED EQUINE ANATOMY

Thoracic Sling

Trapezius

Actions & Functions:

- Both parts together help elevate the shoulder
- The cervical part swings the scapula cranially
- The thoracic part swings the scapula caudally

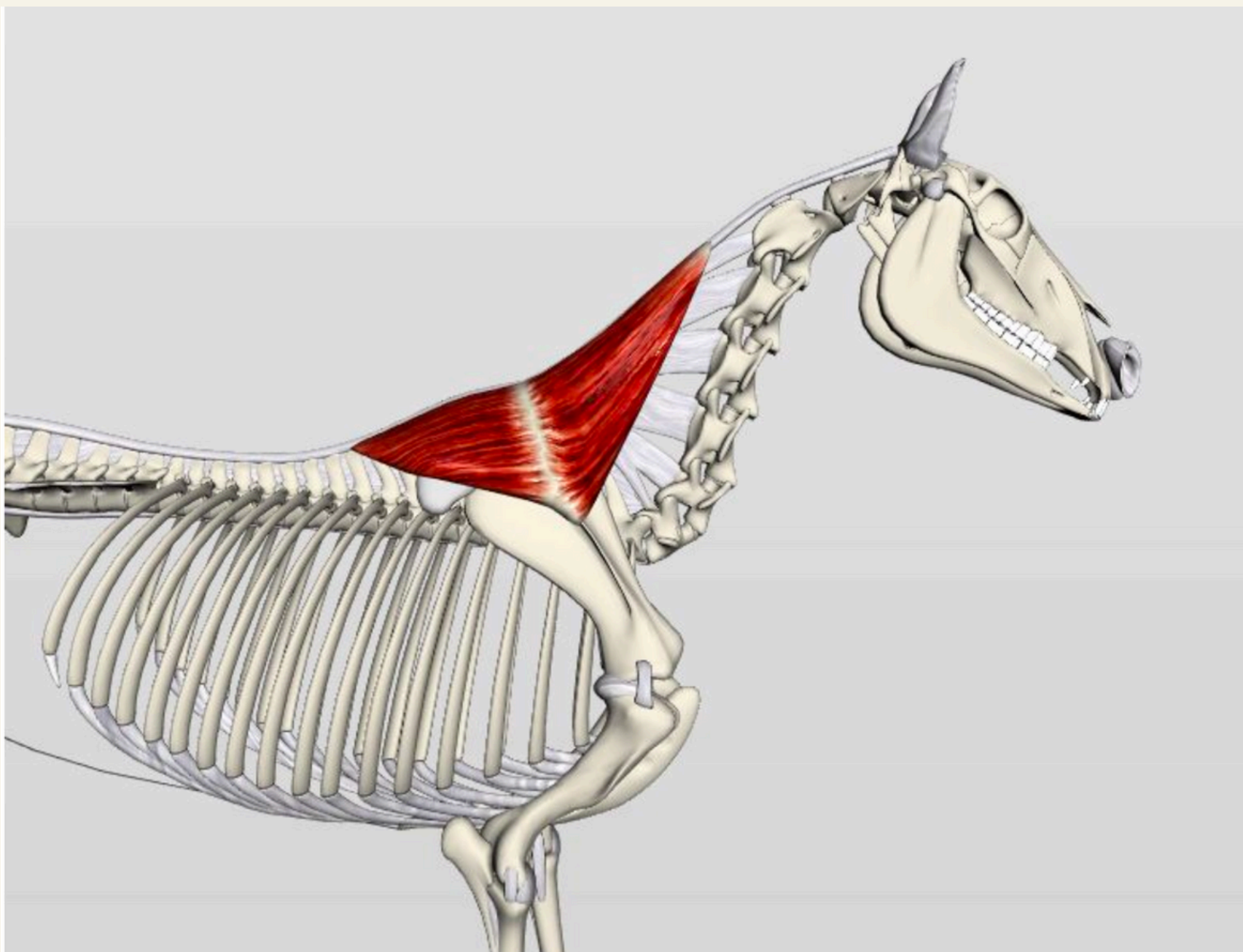
Origin:

- Cervical section: Funicular part of the nuchal ligament
- Thoracic section: Supraspinous ligament of the withers

Insertion: Spine of the scapula (both parts)

Special Notes:

There are two distinct sections — cervical and thoracic. This is one of the most visible and easily palpable superficial muscles in the body.



APPLIED EQUINE ANATOMY

Thoracic Sling

Triceps Brachii

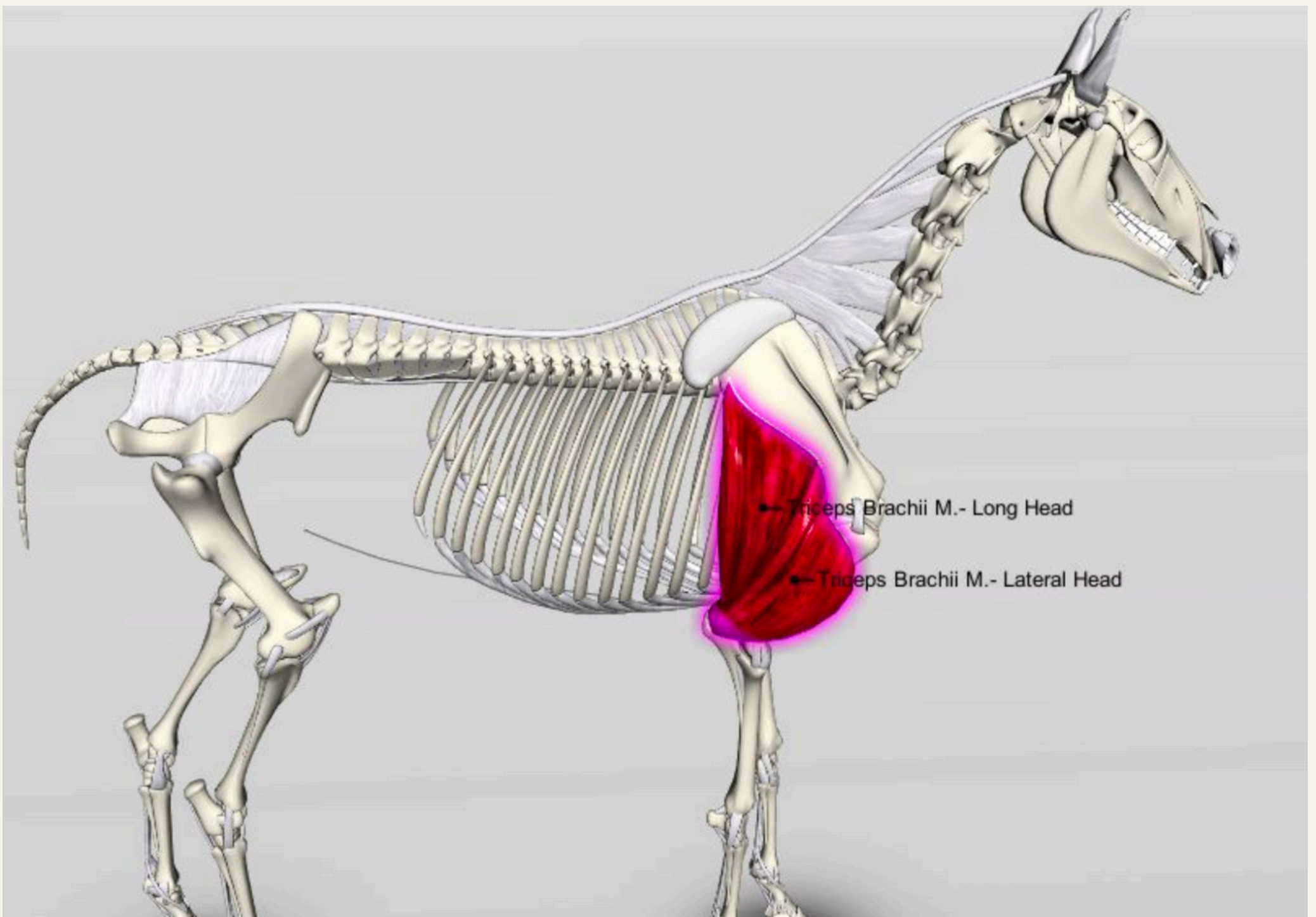
Actions & Functions:

- All three heads extend the elbow joint
- The long head also flexes the shoulder joint
- The long head also plays a role in the stay apparatus

Origin:

- Long head: Caudal border of the scapula
- Lateral head: Deltoid tuberosity on the lateral humerus
- Medial head: Medial surface of the middle third of the humerus

Insertion: Olecranon



APPLIED EQUINE ANATOMY

Thoracic Sling

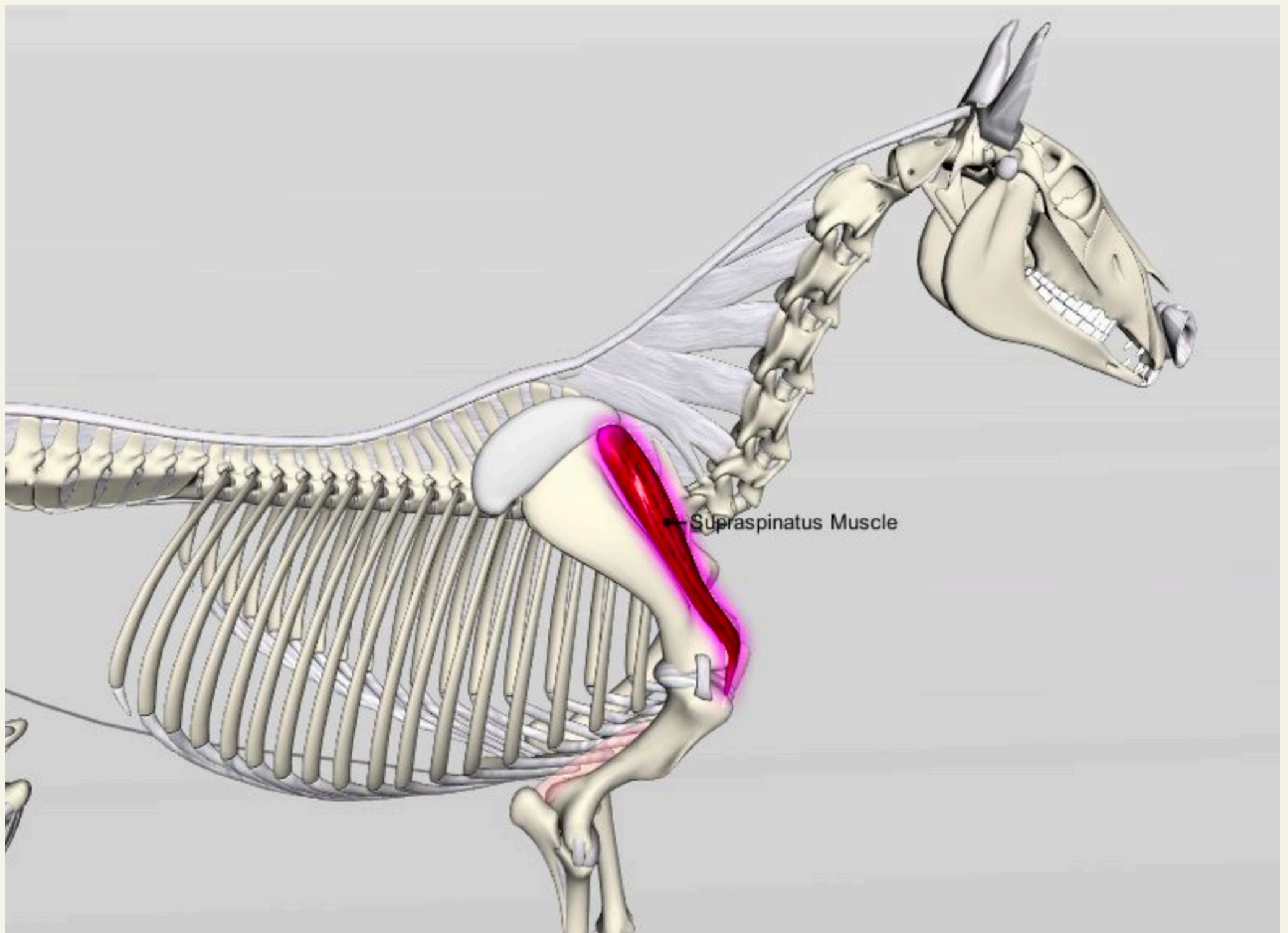
Supraspinatus

Actions & Functions:

- Extends the shoulder joint
- Stabilizes the shoulder against shock and strain

Origin: Supraspinous fossa on the lateral scapula

Insertion: Cranial part of the greater and lesser tubercles of the humerus



APPLIED EQUINE ANATOMY

Thoracic Sling

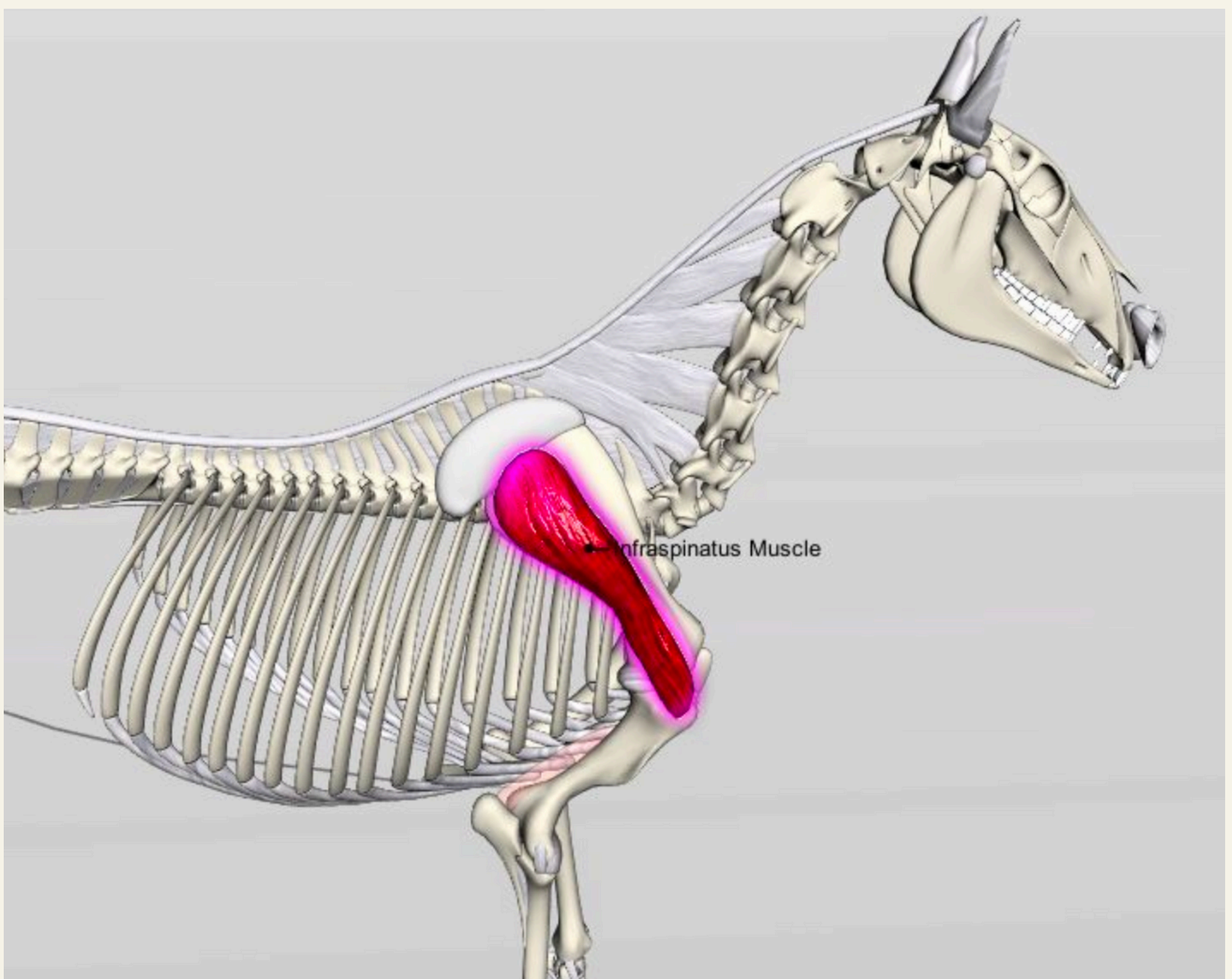
Infraspinatus

Actions & Functions:

- Abducts and laterally rotates the humerus
- Along with the supraspinatus, supports and stabilizes the shoulder joint against shock

Origin: Infraspinous fossa of the lateral scapula

Insertion: Cranial part of the greater tubercle of the humerus



APPLIED EQUINE ANATOMY

Thoracic Sling

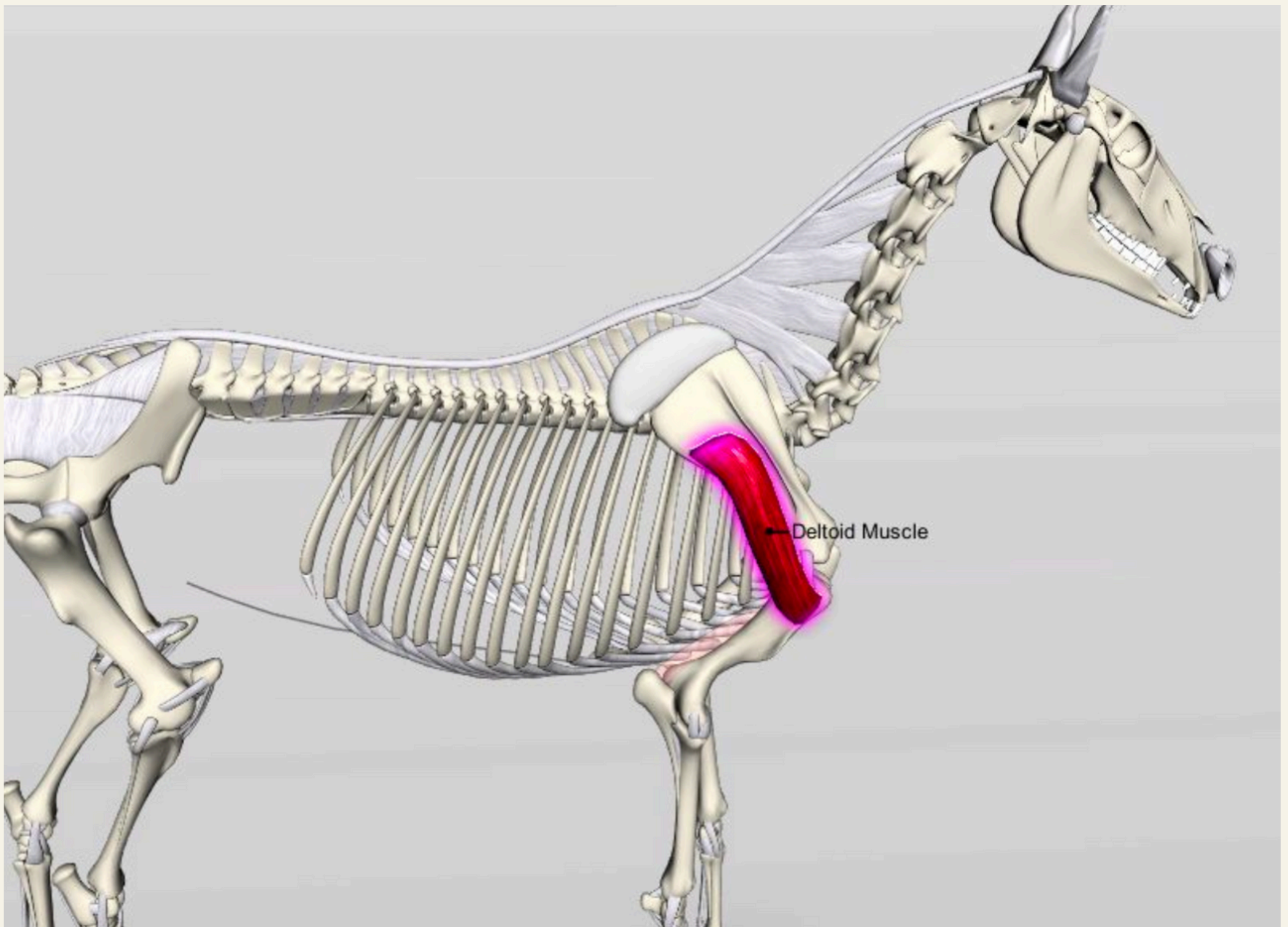
Deltoid

Actions & Functions: Flexes the shoulder joint and abducts the limb.

Abducts humerus

Origin: Caudal border of the spine of the scapula

Insertion: Deltoid tuberosity



APPLIED EQUINE ANATOMY

Thoracic Sling

Subscapularis

Actions & Functions:

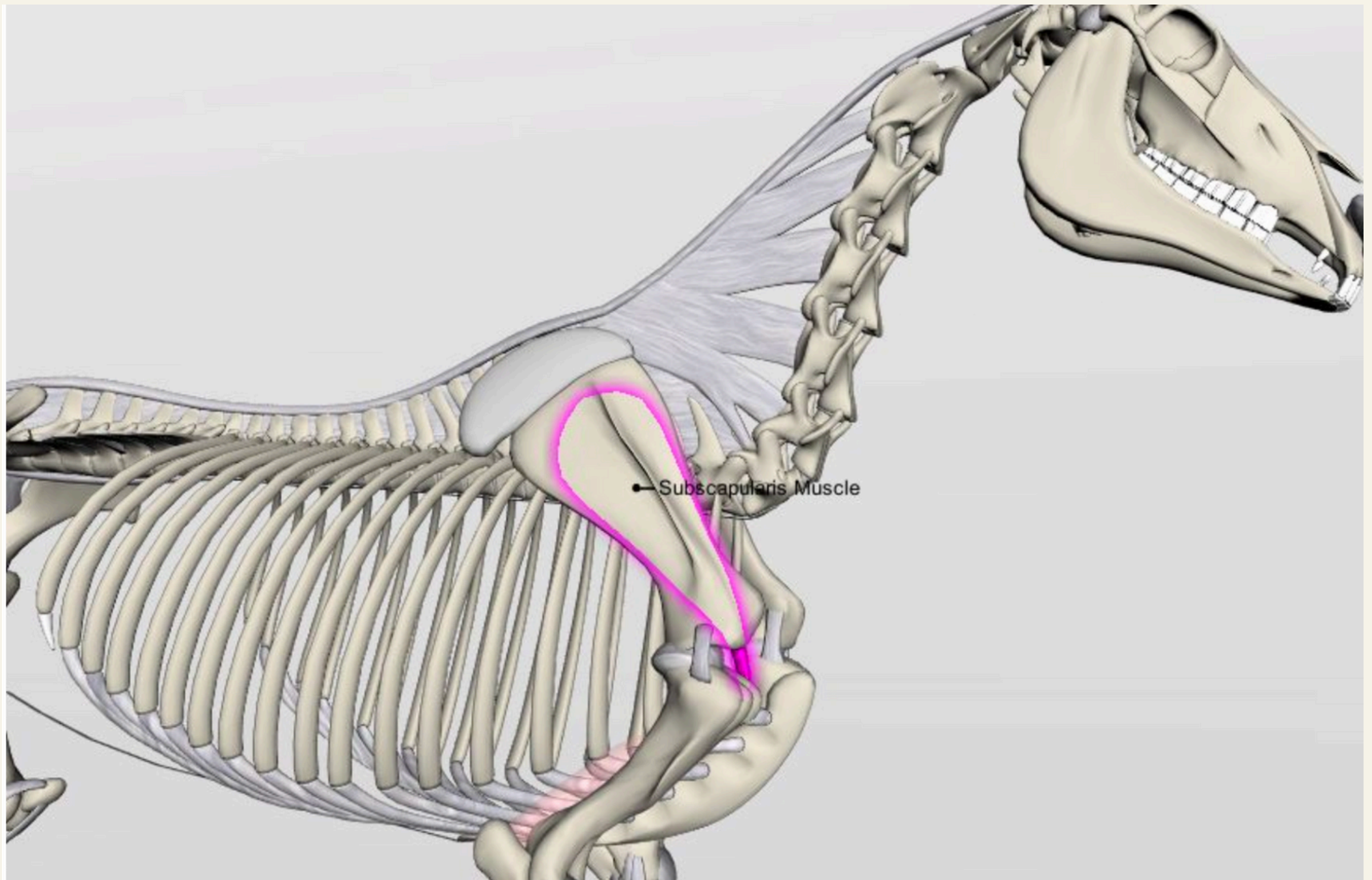
- Adducts the humerus
- Along with the supraspinatus and infraspinatus, helps stabilize the shoulder joint
- Acts as a secondary adductor of the shoulder

Origin: Subscapular fossa

Insertion: Caudal part of the lesser tuberosity of the humerus

Special Notes:

Note the pink outline indicates that this tissue is on the medial aspect of the bone.



APPLIED EQUINE ANATOMY

Thoracic Sling

Teres Major

Actions & Functions:

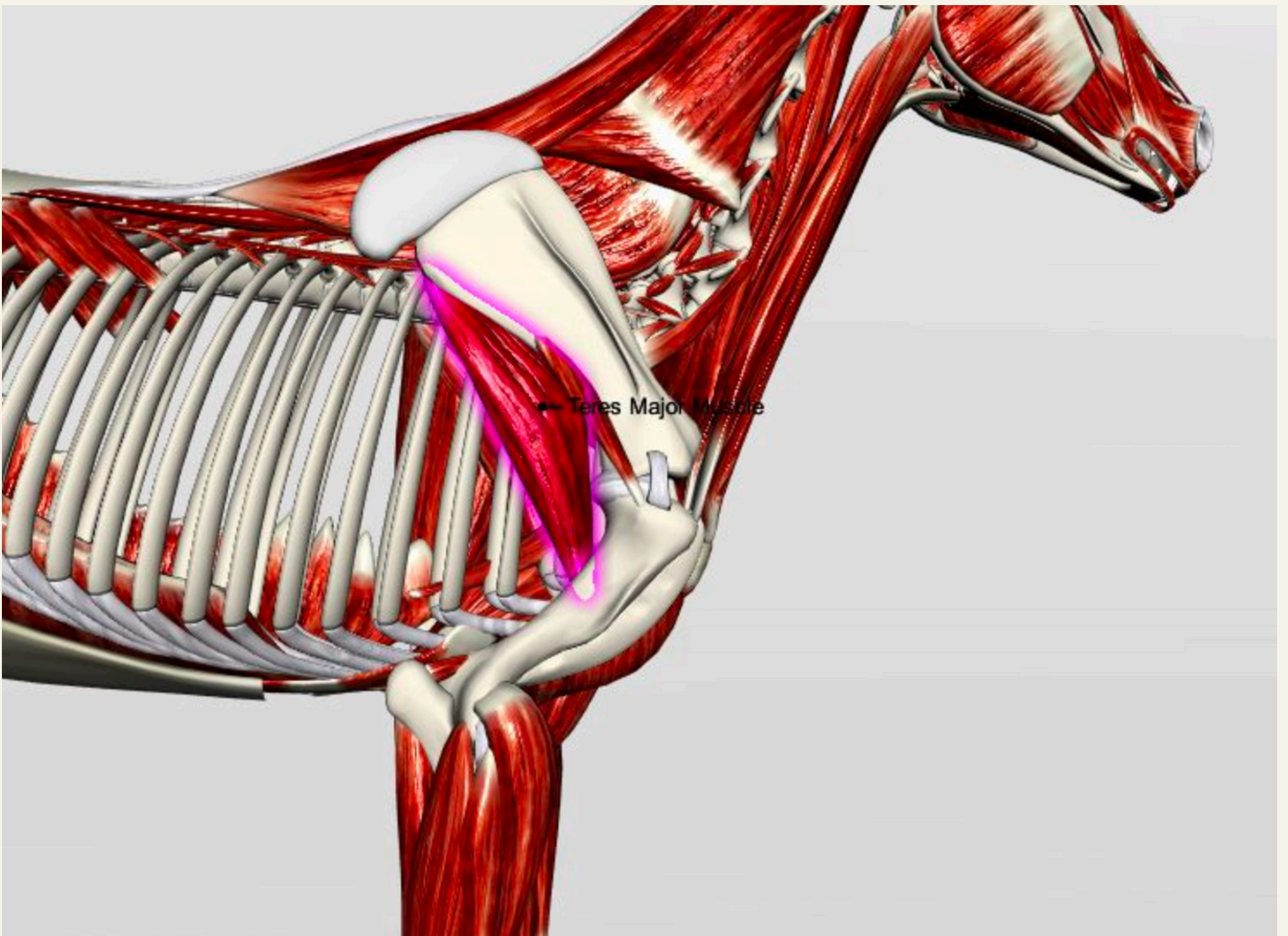
- Flexes the shoulder joint
- Rotates the shoulder medially

Origin: Caudal border of the scapula and the subscapularis muscle

Insertion: Teres major tuberosity of the humerus

Special Notes:

This muscle works together with the latissimus dorsi in strong flexion of the shoulder.



APPLIED EQUINE ANATOMY

Thoracic Sling

Teres Minor

Actions & Functions:

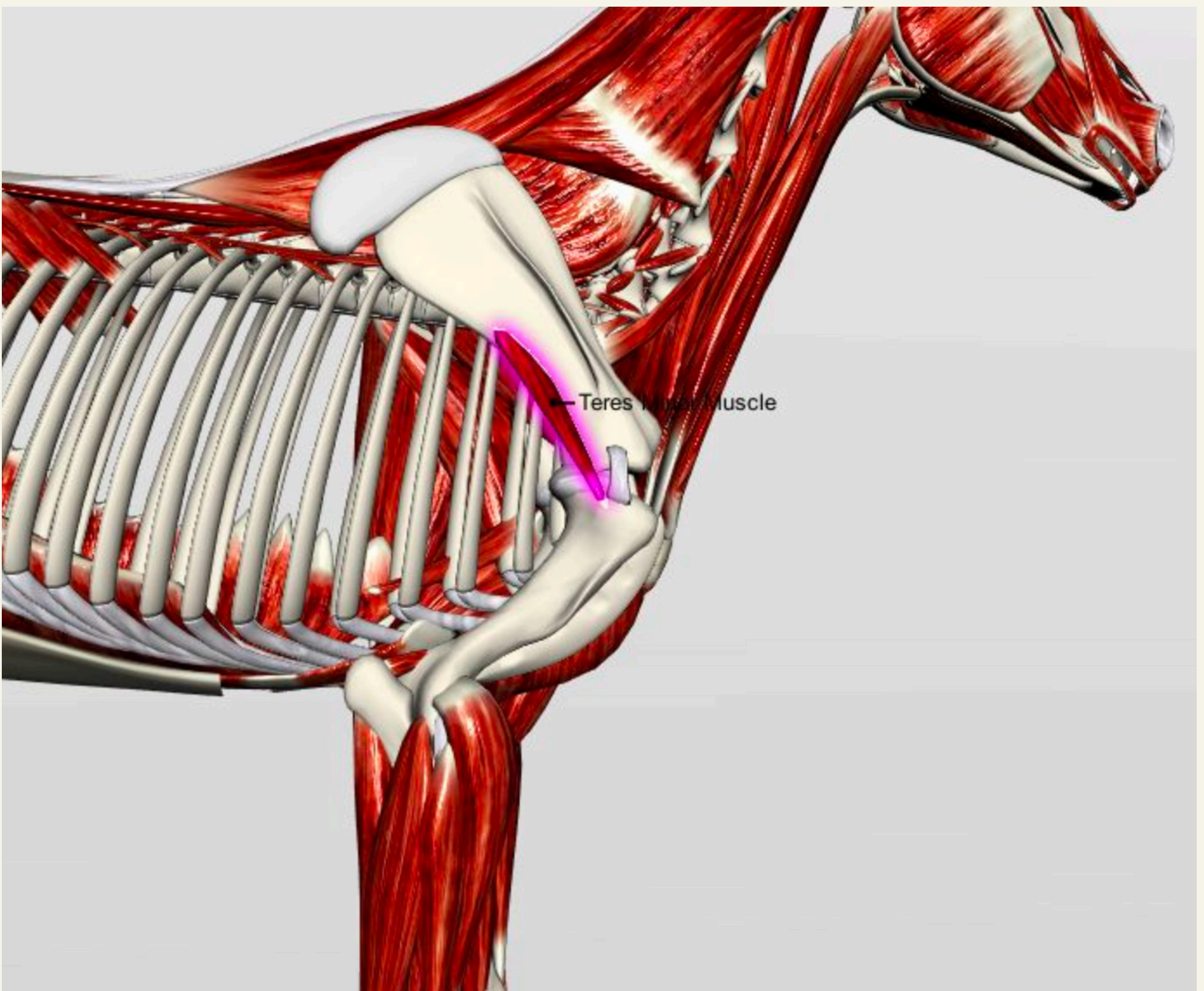
- Flexes the shoulder joint
- Abducts the limb

Origin: Caudal border of the scapula

Insertion: Deltoid and teres minor tuberosity

Special Notes:

This muscle works in conjunction with the infraspinatus.



APPLIED EQUINE ANATOMY

Thoracic Sling

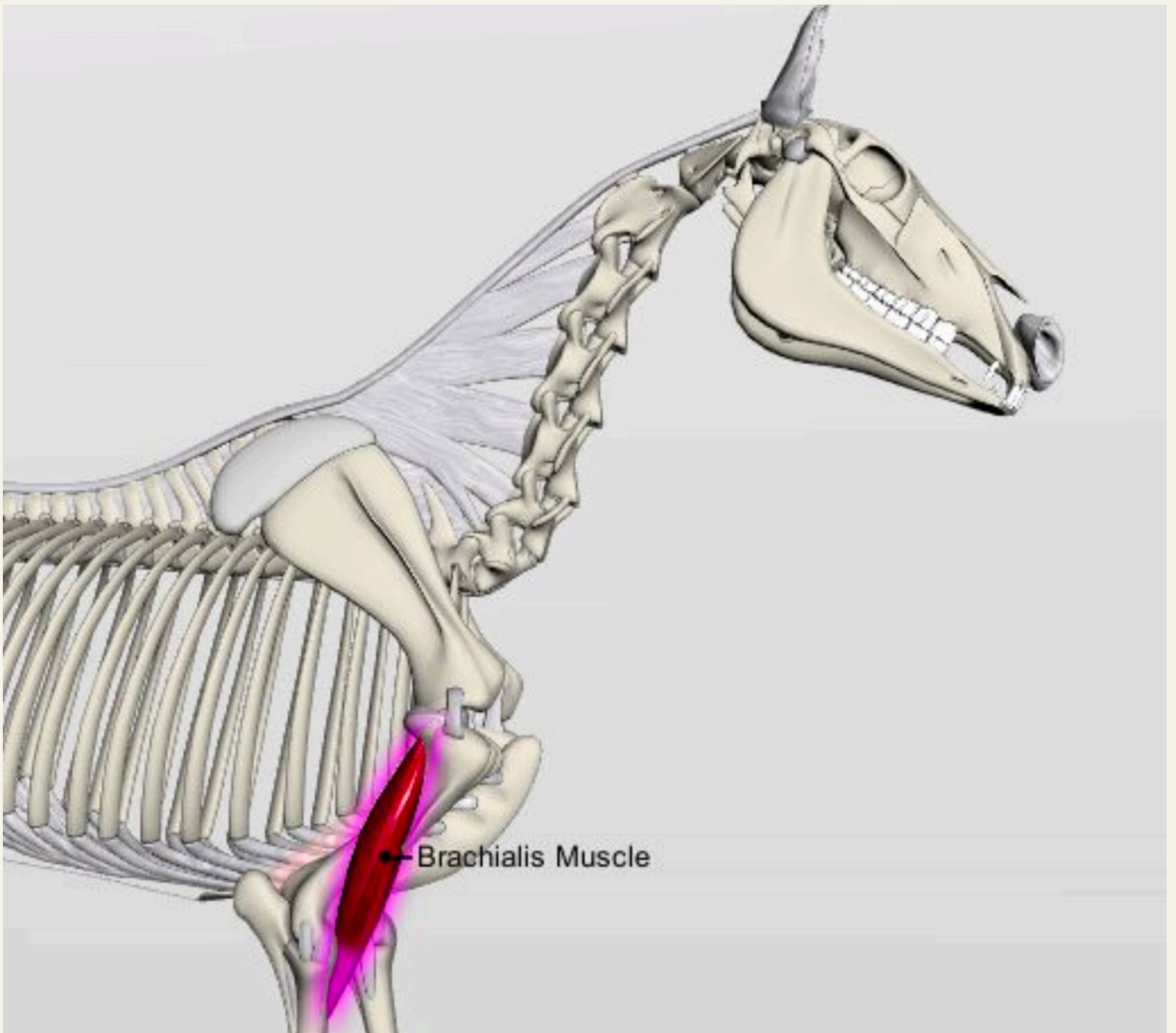
Brachialis

Actions & Functions:

- Flexes the elbow joint
- Laterally rotates the radius

Origin: Proximal third of the caudal aspect of the humerus

Insertion: Proximal medial aspect of the radius



APPLIED EQUINE ANATOMY

Thoracic Sling

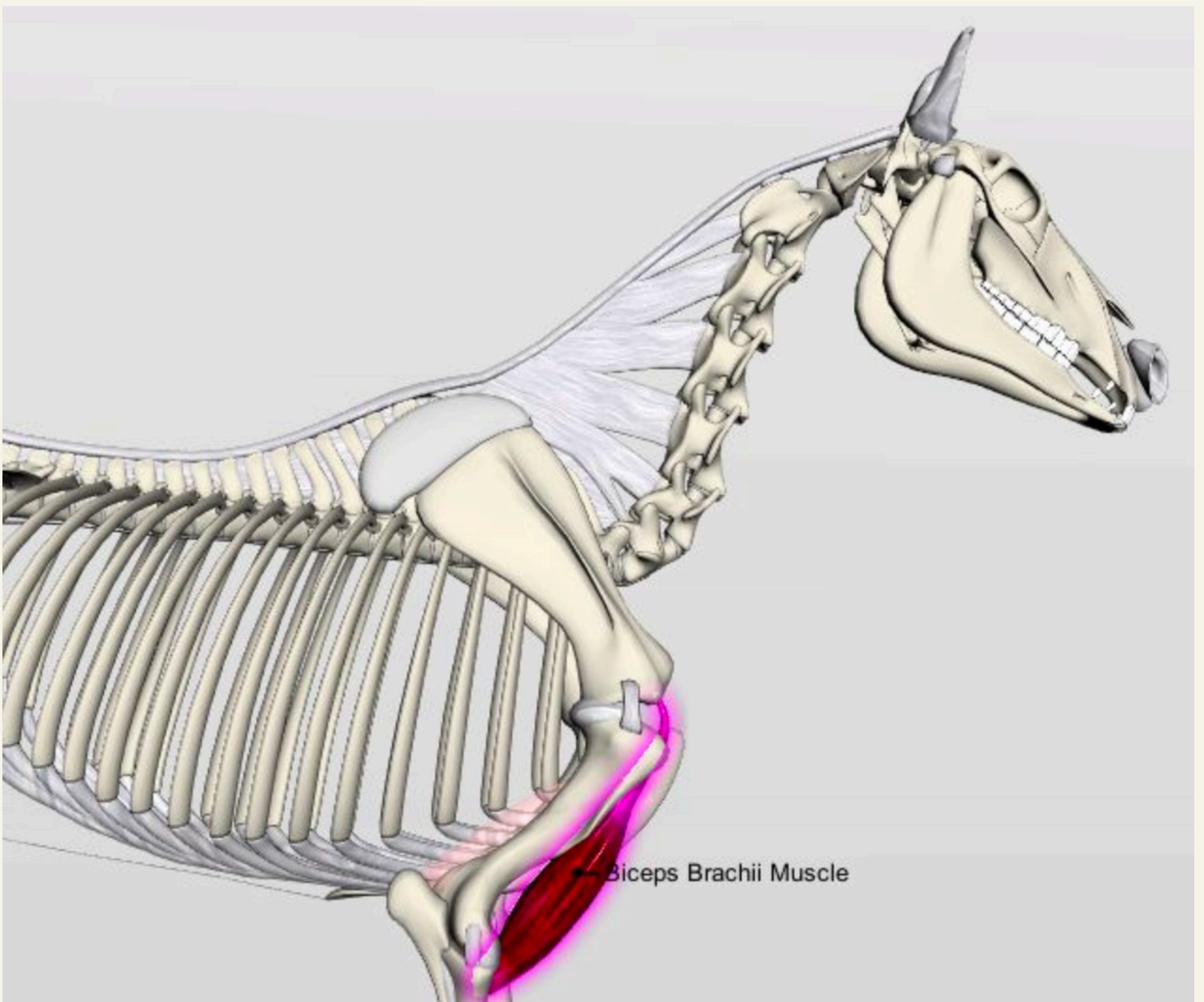
Biceps brachii

Actions & Functions:

- Extends the shoulder joint
- Flexes the elbow joint
- Primary fixator of the shoulder joint
- Supports the carpus via the lacertus fibrosus

Origin: Supraglenoid tubercle of the scapula

Insertion: Radial tuberosity; medial collateral ligament of the elbow; radial carpal extensor via the lacertus fibrosus



APPLIED EQUINE ANATOMY

Thoracic Sling

Transverse Pectoralis

Actions & Functions:

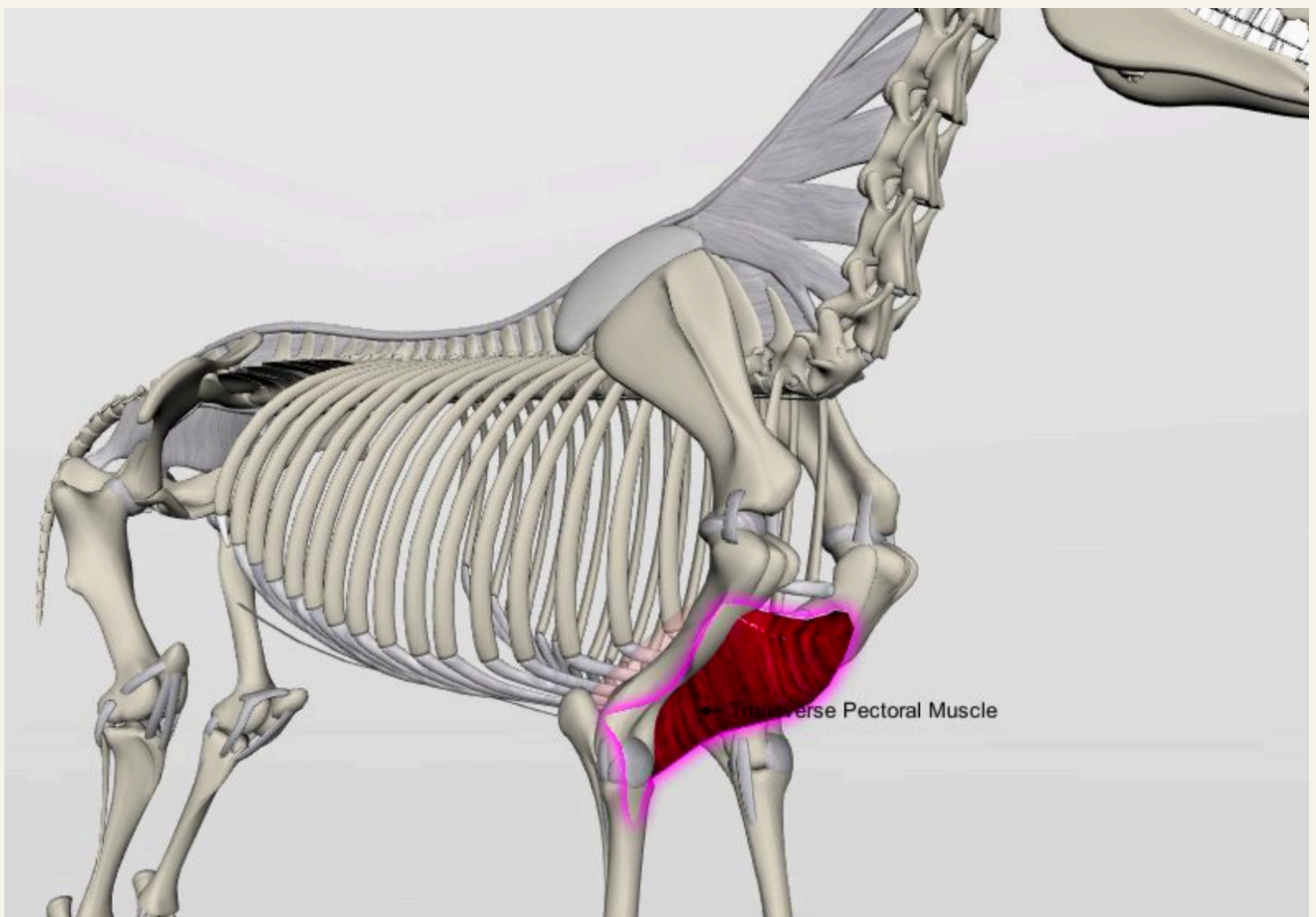
- Adducts the forelimb

Origin: Cranial sternum

Insertion: Crest of the humerus and fascia of the forearm

Special Notes:

This is one of the two superficial pectoral muscles.



APPLIED EQUINE ANATOMY

Thoracic Sling

Descending Pectoral (Cranial Superficial Pectoral)

Actions & Functions:

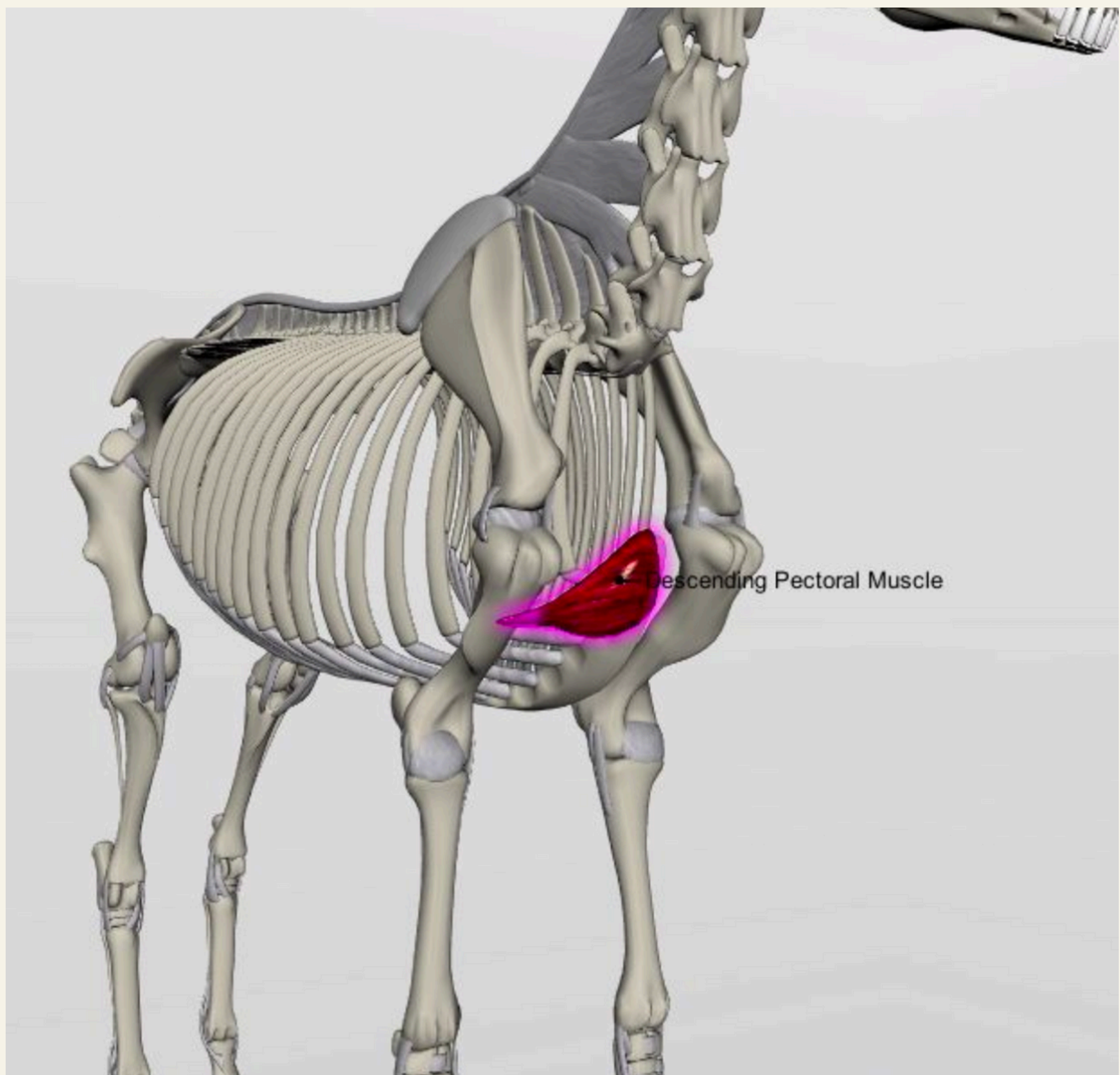
- Adducts and advances the forelimb

Origin: Manubrium

Insertion: Humeral crest and fascia of the arm

Special Notes:

This is one of the two superficial pectoral muscles.



APPLIED EQUINE ANATOMY

Thoracic Sling

Caudal Deep Pectoral (Ascending Pectoral)

Actions & Functions:

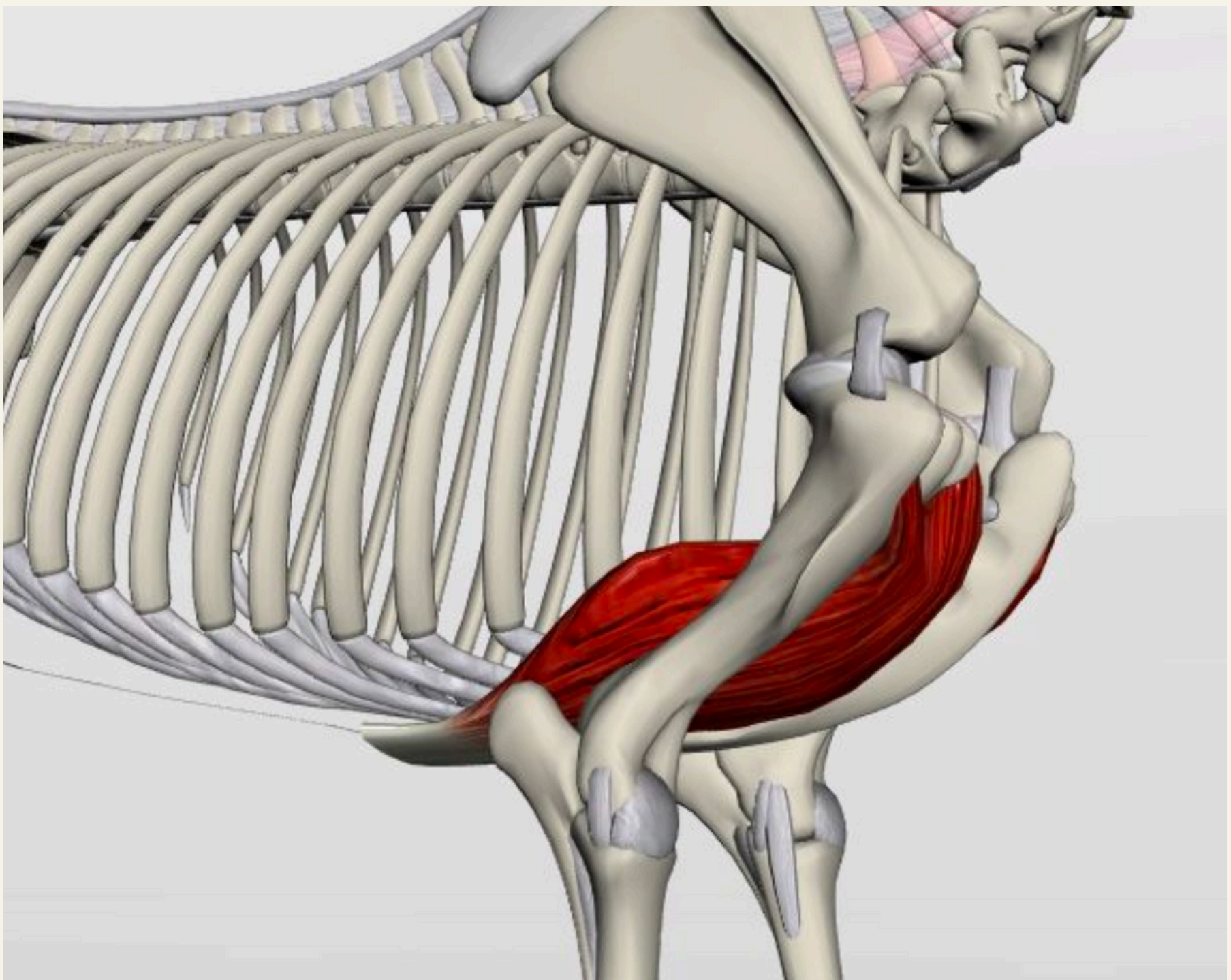
- Adducts the limb

Origin: Caudal part of the sternum

Insertion: Greater and lesser tubercles of the humerus

Special Notes:

This is one of the two deep pectoral muscles. Pain or constriction in this muscle can cause a horse to react negatively to being girthed or cinched. In these cases, the muscle is often tender even to light palpation.



APPLIED EQUINE ANATOMY

Thoracic Sling

Subclavius

Actions & Functions:

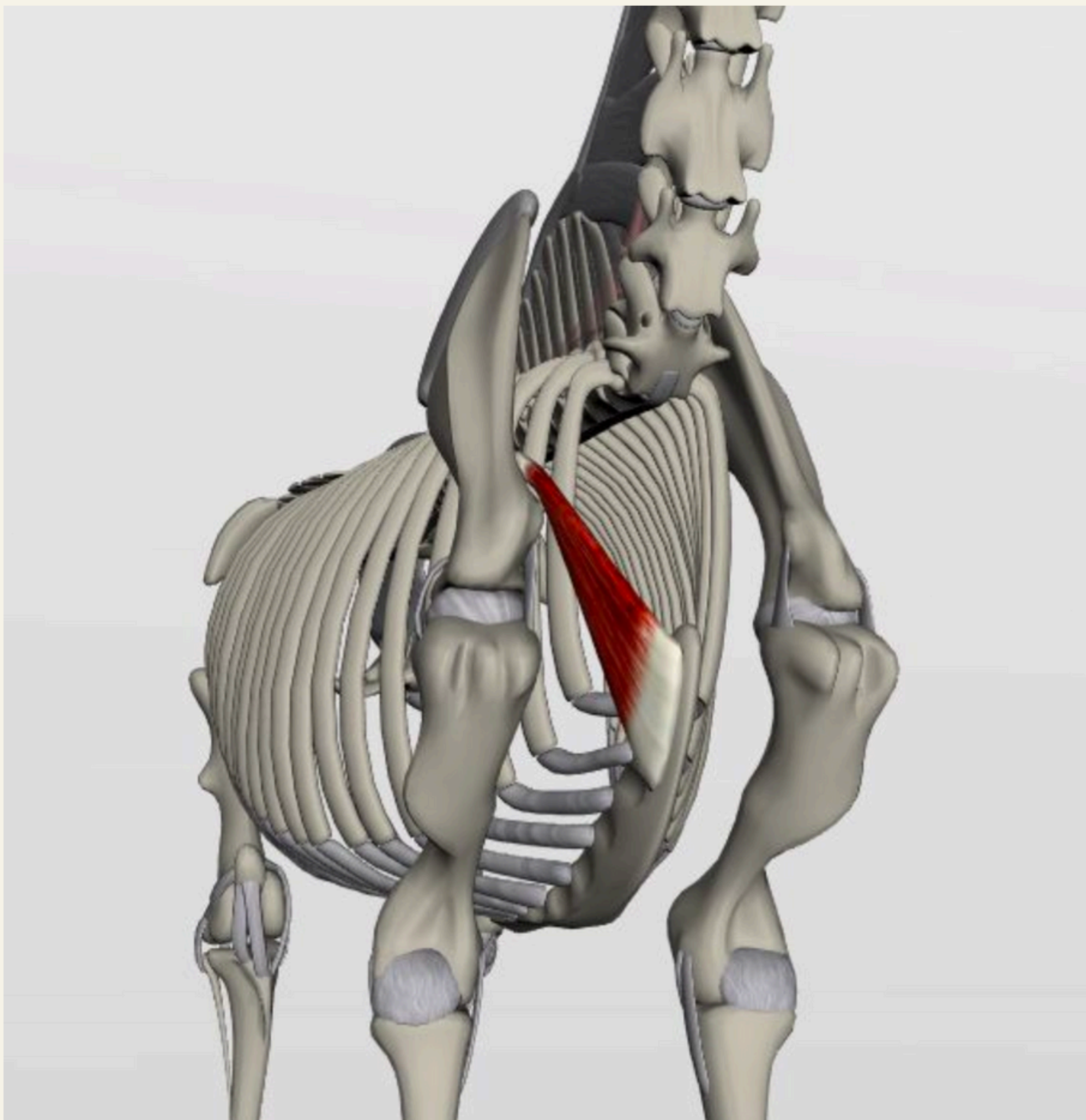
- Adducts the limb

Origin: Cranial part of the sternum

Insertion: Epimysium of the supraspinatus (elastic covering over the muscle)

Special Notes:

This is one of the two deep pectoral muscles.



APPLIED EQUINE ANATOMY

Thoracic Sling

Latissimus Dorsi

Actions & Functions:

- Pulls the barrel forward onto the advancing limb when the limb is on the ground
- Pulls the entire shoulder girdle caudally

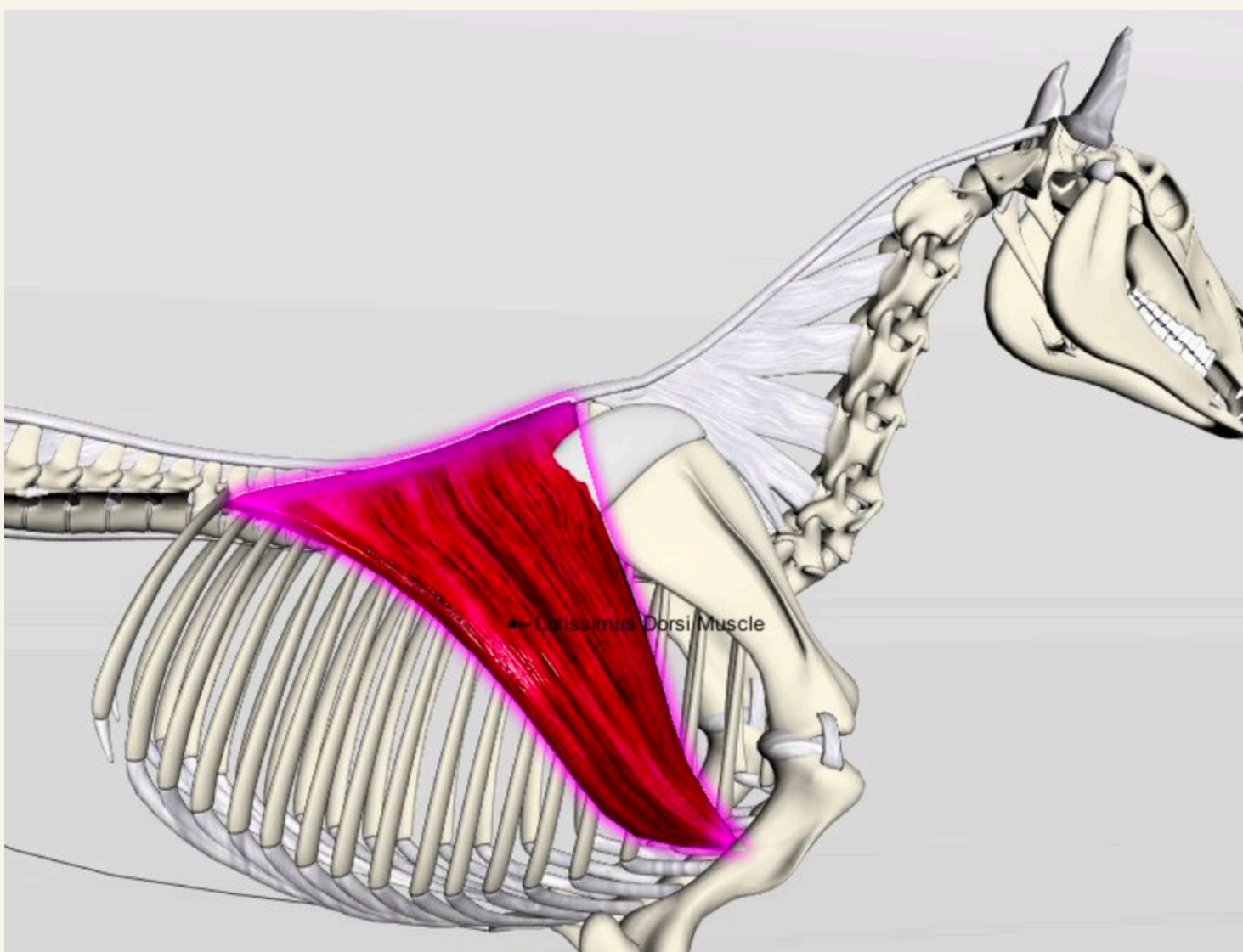
Origin: Supraspinous ligament and thoracolumbar fascia

Insertion: Teres major tuberosity of the humerus

Special Notes:

This muscle is one of the primary muscles to become constricted, even in response to the most sensitive riders, as the girth and saddle can restrict its natural movement. Adhesions often form in the following areas in many horses, regardless of saddle:

- a) Along the caudal border of the muscle where it overlays the ribs
- b) Where the latissimus dorsi emerges from underneath the triceps/deltoid
- c) In the intramuscular septum between the trapezius thoracis and the latissimus dorsi



APPLIED EQUINE ANATOMY

Thoracic Sling

Serratus Ventralis Cervicis

Actions & Functions:

- Supports the weight of the thorax
- When the limb is on the ground, helps raise the lower half of the neck up through the withers

Origin: Transverse processes of C3–C7

Insertion: Serrated medial surface of the scapula and scapular cartilage

Special Notes:

The two serratus ventralis muscles are deep muscles that form the primary inner “thoracic sling” of soft tissue that suspends the barrel and chest between the shoulders. These muscles are hugely important for the lift in the chest and withers that the rider feels when the horse is truly “on the bit” and moving in correct biomechanical form. When the horse’s forward action is impeded—due to overuse of the reins, poor saddle fit, or other factors—these muscles can become tight, constricted, and painful, making it increasingly difficult for the horse to perform correctly.

Serratus Ventralis Thoracis

Actions & Functions:

- Supports the weight of the thorax
- When the limb is on the ground, helps raise the trunk up through the withers

Origin: The first eight ribs

Insertion: Serrated medial surface of the scapula and scapular cartilage

Special Notes:

Of the two serratus ventralis muscles, the thoracic section is especially prone to inhibition of natural movement—no matter how careful the rider—because the caudal half of this muscle lies under the girth while the horse is being ridden.

APPLIED EQUINE ANATOMY

Thoracic Sling

